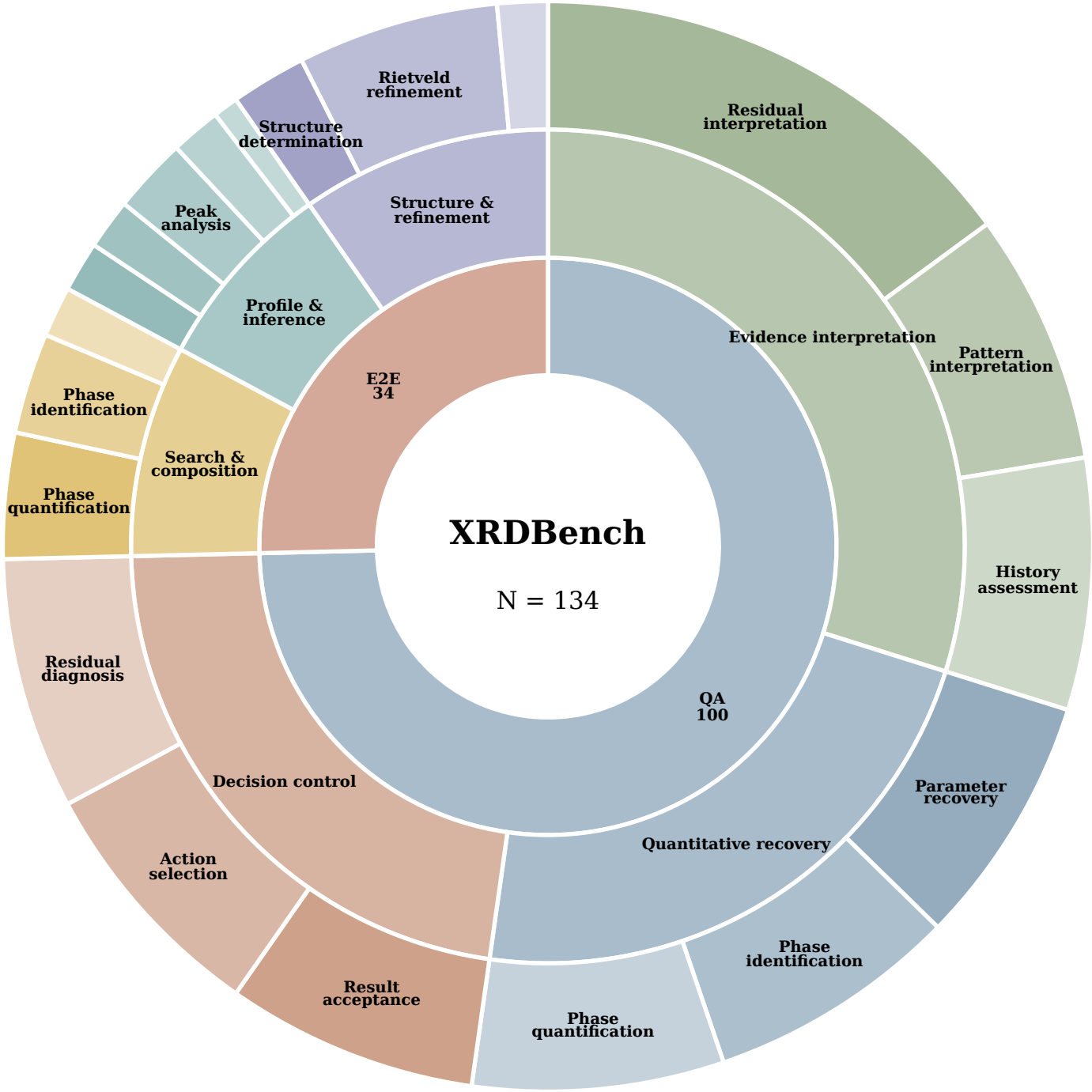




Inner Ring: Evaluation Track

- XRDBench-QA: 100 tasks
- XRDBench-E2E: 34 tasks

Difficulty

QA: 30 Easy / 40 Medium / 30 Hard

E2E: 11 Medium / 23 Hard

Middle Ring: Capability Stratum

- QA - Evidence interpretation: 40
- QA - Quantitative recovery: 30
- QA - Decision control: 30
- E2E - Search & composition: 11
- E2E - Profile & inference: 10
- E2E - Structure & refinement: 13

Outer Ring: Task Family

XRDBench-QA

- History assessment: 10
- Pattern interpretation: 10
- Residual interpretation: 20
- Parameter recovery: 10
- QA phase identification: 10
- QA phase quantification: 10
- Action selection: 10
- Residual diagnosis: 10
- Result acceptance: 10

XRDBench-E2E

- Database Retrieval: 2
- Phase Identification: 4
- Phase Quantification: 5
- Crystallite Analysis: 2
- Defect Disorder: 2
- Material Property Prediction: 1
- Pattern Prediction: 2
- Peak Analysis: 3
- Indexing: 2
- Rietveld Refinement: 8
- Structure Determination: 3

Ring areas encode task counts.